

AMENDMENT NO. 1 NOVEMBER 1980
TO
IS : 7803 (Part I) - 1975 SPECIFICATION FOR
PLASTIC CONTAINERS FOR PHARMA-
CEUTICAL USE

PART I OTHER THAN PARENTERAL AND OPHTHALMIC
PREPARATIONS

Alterations

(*First cover, pages 1 and 3, title*) — Substitute the following for the existing title:

‘ Indian Standard

SPECIFICATION FOR POLYETHYLENE
CONTAINERS FOR PHARMACEUTICAL USE

PART I OTHER THAN PARENTERAL AND OPHTHALMIC
PREPARATIONS

(*Page 3, clause 0.2*) — Substitute the following for the existing clause:

‘0.2 With the growth of indigenous packaging industry, polyethylene containers are replacing conventional containers made of materials, such as glass, metal, etc. Polyethylene containers are also being increasingly used for pharmaceutical trade.’

(*Page 3, clause 1.1 line 1*) — Substitute ‘ polyethylene ’ for ‘ plastic ’.

(*Page 4, clause 3.1*) — Substitute the following for the existing clause:

‘3.1 Material — Only virgin polyethylene material conforming to the requirements of IS : 7277-1974† which is practically odourless, shall be used in the manufacture of the containers or any accessory product. Pigments used, if any, shall be those which are approved for the purpose of colouring plastics and are approved for applications involving contact with food and pharmaceuticals.’

NOTE — Specifications for pigments to be used in contact with food and pharmaceuticals are presently under preparation and these pigments shall conform to the requirements of such future specifications.

Addenda

(*Page 4, clause 4.3.4*) — Add the following new clause after **4.3.4**:

‘4.4 Compatibility Test — The compatibility tests shall be carried out in order to establish the compatibility of the container with the contents so that the quality, purity and strength of the active ingredients of pharmaceuticals when packed in polyethylene containers are ensured. The test shall be carried out as given in Appendix H.’

(*Page 4, foot-note with ‘*’ mark*) — Add the following new foot-note after ‘*’ mark:

‘†Code of practice for safe use of polyethylene in contact with foodstuffs, pharmaceutical and drinking water.’

(*Page 11, clause G-3.1*) — Add the following new appendix after **G-3.1**:

‘APPENDIX H

(*Clause 4.4*)

METHOD OF TEST FOR COMPATIBILITY OF POLYETHYLENE CONTAINERS

H-1. PRINCIPLE

H-1.1 Pieces of polyethylene material with which the container is made are treated at an elevated temperature with the liquid which the container is intended to contain. The compatibility tests shall be carried out in accordance with **H-3.1** and **H-3.2** if the contents to be packed are liquid preparations. In case of dry powdered pharmaceuticals the method given in **H-3.2** only shall be applicable.

H-2. TEST SPECIMEN

H-2.1 Material — Three test pieces of approximately equal size and having an area of not less than 15 cm² shall be cut from any convenient part of the container. Each test piece shall be cleaned, wiped and dried. It shall be measured for length, width and thickness to the nearest 0.05 mm and weighed to the nearest milligram.

H-2.2 Container — Six samples of specific container intended for packing of particular product shall be tested in accordance with the procedure given in **H-3.2**.

H-3. PROCEDURE

H-3.1 Testing of Material — The liquid which is intended to be filled in the container shall be introduced into a glass vessel and test pieces completely immersed, avoiding unnecessary contact with the other pieces or the walls of the glass vessel. Where the density of plastics material is less than that of the liquid, small weights inert to the liquid, may be used to prevent the test pieces from either floating or curling. The test shall be carried out continuously over 28 days at a temperature of $50 \pm 2^\circ\text{C}$. The liquid and the test pieces shall be thoroughly agitated every 24 hours.

After the required test period has elapsed, the test pieces shall be removed from the liquid, suitably cleaned, dried, weighed and measured.

H-3.2 Testing of Container — In order to assess the compatibility of the container, the container shall be filled with the product to the nominal capacity, sealed and capped in the manner intended and tested at any one of the following conditions, depending upon the thermolabile characteristic of the product being tested:

- a) $50 \pm 2^\circ\text{C}$ for a period of 28 days,
- b) $37 \pm 2^\circ\text{C}$ for a period of 3 months,
- c) $27 \pm 2^\circ\text{C}$ for 1/3 of the anticipated shelf life period, and
- d) $37 \pm 2^\circ\text{C}$ and 75 ± 5 percent relative humidity for 3 months (for preparations other than liquids).

H-3.2.1 The above test conditions are intended to serve as a guideline for usage of the container. However, a shelf life storage test at $27 \pm 2^\circ\text{C}$ or at ambient temperature should be carried out till the anticipated shelf life period of the product in order to confirm the suitability of the container.

H-4. TEST RESULTS AND INTERPRETATION

H-4.1 Material — At the end of the test period any change in weight, dimension or alterations in other characteristics, such as colour, blooming, etc, shall be used for determining the suitability of the container for the intended purpose.

H-4.2 Containers — At the end of the test period, containers shall be examined for the following:

- a) Visible cracks, if any;
- b) Change in colour;

- c) Change in weight; and
- d) Change in shape.

H-4.2.1 In the event of any sign of crack, the container shall be deemed to be unsuitable for use for the particular product. Any change in weight or shape or both shall be used by the manufacturer and purchaser to determine the suitability of the container. A very slight change in colour shall be considered insignificant.

H-4.2.2 In addition, the contents shall be tested to ensure that it meets the house specification of the pharmaceutical manufacturer and shall be substantially free from foreign odour and taste. '

(MCPD 11)